Clustering Spotify Tracks

Mood and Rhythm Analysis Using Unsupervised Learning

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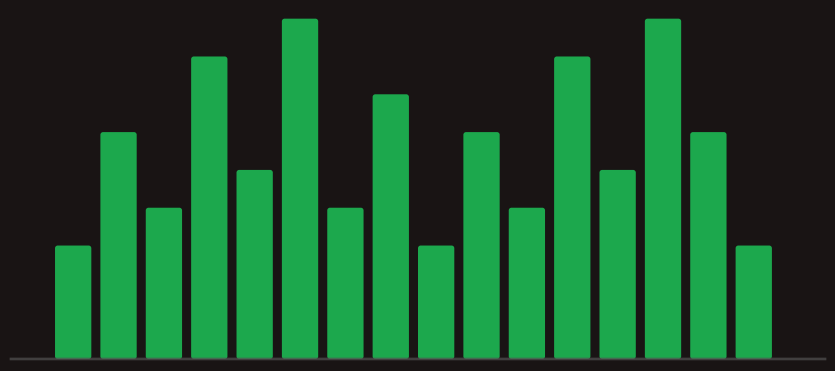
January 10, 2026

Abstract

This project applies unsupervised learning techniques to Spotify audio features in order to uncover latent structure in musical moods and rhythmic patterns. Using clustering algorithms such as Hierarchical Clustering, Gaussian Mixture Models (GMM), and DBSCAN, we explore how tracks group according to emotional intensity, positivity, and groove characteristics. The results demonstrate that meaningful and interpretable clusters emerge from audio features alone, offering insight into music recommendation, genre discovery, and mood-based playlist generation.



Data • Rhythm • Mood • Structure

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1 Introduction

With the rapid growth of music streaming platforms, understanding the structure of musical content has become increasingly important. Spotify, one of the world's largest music streaming services, provides a rich collection of audio features that describe musical, rhythmic, and emotional properties of tracks. These features make it possible to analyze music in a data-driven way, beyond traditional genre labels.

The goal of this project is to explore whether meaningful patterns can be discovered in music using **unsupervised learning techniques**. Instead of relying on predefined genres or human annotations, clustering algorithms are applied directly to Spotify audio features to uncover latent structures related to musical mood, energy, and rhythm.

In particular, this project investigates:

- Whether songs naturally group into distinct **mood-based clusters** using emotional features such as valence and energy
- How **rhythmic patterns** form dense regions in feature space, and where atypical or experimental tracks appear as outliers
- The strengths and limitations of different clustering algorithms when applied to musical data

By applying Hierarchical Clustering, Gaussian Mixture Models (GMM), and DBSCAN, this study shows how unsupervised learning can reveal interpretable musical structures and provide useful insights for music recommendation systems, playlist generation, and exploratory music analysis.

2 Dataset Description

2.1 Overview

The dataset used in this project consists of Spotify tracks spanning **125 different music genres**. Each track is associated with a set of audio features extracted using Spotify's Audio Features API. The data is provided in CSV format, making it efficient to load and process for analysis.

This dataset is well-suited for unsupervised learning tasks due to its rich numerical feature space and diversity of musical styles.

2.2 Dataset Usage

The dataset can be used for a variety of music-related data science applications, including:

- Building music recommendation systems based on user preferences
- Classification and prediction tasks using audio features
- Exploratory analysis of musical structure and listener behavior

In this project, the dataset is primarily used for **clustering analysis** to identify latent musical patterns without relying on genre labels.

2.3 Feature Description

Each track in the dataset is associated with a set of metadata attributes and numerical audio features extracted using Spotify's Audio Features API. These variables describe musical structure, rhythm, production style, and emotional tone.

For clarity, the full list of available attributes is reported below. However, only a subset of numerical audio features was used for clustering, depending on the specific analytical goal (mood, rhythm, texture, or subspace structure).

- **track_id**: Unique Spotify identifier for the track
- **artists**: Names of the performing artists
- **album_name**: Album in which the track appears
- **track_name**: Name of the track
- **popularity**: A value between 0 and 100 indicating the track's popularity
- **duration_ms**: Track length in milliseconds
- **explicit**: Indicates whether the track contains explicit lyrics
- **danceability**: Measure of how suitable the track is for dancing (0.0–1.0)
- **energy**: Measure of intensity and activity (0.0–1.0)
- **key**: Musical key of the track using pitch class notation
- **loudness**: Overall loudness in decibels (dB)
- **mode**: Indicates whether the track is in a major (1) or minor (0) mode
- **speechiness**: Presence of spoken words in the track
- **acousticness**: Confidence measure of whether the track is acoustic
- **instrumentalness**: Likelihood that the track contains no vocals
- **liveness**: Probability that the track was recorded live
- **valence**: Musical positiveness conveyed by the track
- **tempo**: Estimated tempo in beats per minute (BPM)
- **time_signature**: Estimated time signature of the track
- **track_genre**: Genre label associated with the track

From this set, only numerical audio features were used for clustering. Feature selection varied across analyses:

- **Mood analysis**: valence, energy, danceability
- **Rhythm analysis**: tempo, energy, danceability
- **Texture analysis**: acousticness, instrumentalness, loudness
- **Subspace analysis**: tempo, energy, valence

Non-numerical variables such as track identifiers, artist names, album names, and genre labels were excluded from the clustering process and retained only for interpretation and validation of the results.

2.4 Data Preprocessing

The preprocessing phase focused on ensuring data quality and preparing the dataset for clustering analysis. The following steps were performed:

1. **Dataset Inspection** The size and structure of the dataset were examined by checking its dimensions and column names. This step confirmed the number of tracks and available features and verified that the data had been loaded correctly.

2. **Duplicate Detection** Duplicate rows were searched for in order to identify repeated observations. Because duplicate tracks can artificially increase the density of certain regions in the feature space, removing them is important for fair clustering.

3. **Missing Value Assessment** All selected features were checked for missing values. Since most clustering algorithms cannot handle missing data directly, this step ensured that no imputation or row removal was required before analysis.

4. **Feature Selection** Only numerical audio features were retained for clustering, as distance-based and probabilistic methods require numerical input. Non-numerical variables such as track identifiers, artist names, album names, and genre labels were excluded and kept only for interpretation.

5. **Feature Scaling** Numerical features were standardized to have zero mean and unit variance. This was necessary because the features operate on different scales (for example, tempo in BPM versus valence between 0 and 1). Without scaling, features with larger numeric ranges would dominate the clustering process.

6. **Sampling Strategy** The dataset was used without heavy downsampling in order to preserve the diversity of musical styles. Sampling decisions were made conservatively, balancing computational feasibility with the need to retain representative musical patterns.

3 Methodology

3.1 Mood Clustering: Identifying Emotional Profiles

Music expresses emotions in many different ways, and these emotions change gradually rather than falling into strict categories. Still, people often describe music using familiar mood labels such as calm, energetic, or sad.

In this analysis, we examine whether these commonly perceived emotional moods can be identified directly from Spotify audio features, without using genre labels or manual annotations.

The main question guiding this analysis is:

Can songs be grouped into a small number of clear emotional moods based only on their audio characteristics?

3.1.1 Feature Selection

Three audio features were selected to represent emotional characteristics:

- **Valence**: captures the perceived positivity or negativity of a track
- **Energy**: reflects intensity and activity level
- **Danceability**: represents rhythmic suitability for movement

These features jointly provide an interpretable representation of musical emotion, balancing expressive power with visual clarity.

3.1.2 Clustering Method

Agglomerative Hierarchical Clustering was employed to explore the structure of emotional similarities between tracks. This method was chosen because it reveals nested relationships and allows examination of how emotional groupings gradually merge as similarity thresholds change.

Unlike flat clustering approaches, hierarchical clustering aligns well with emotional analysis, as it reflects how subtle emotional differences can branch into broader mood categories.

Mathematical Formulation Hierarchical clustering starts with each track as its own cluster and iteratively merges the two closest clusters. Distances between tracks are measured using Euclidean distance:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^{p} (x_{ik} - x_{jk})^2}$$

Using Ward's linkage, clusters are merged by minimizing the increase in within-cluster variance:

$$\Delta(C_a, C_b) = \frac{|C_a||C_b|}{|C_a| + |C_b|} \|\mu_a - \mu_b\|^2$$

where μ_a and μ_b are the centroids of clusters C_a and C_b.

3.1.3 Visualization and Results

To analyze the hierarchical structure of emotional similarity, a dendrogram was constructed using standardized feature values.

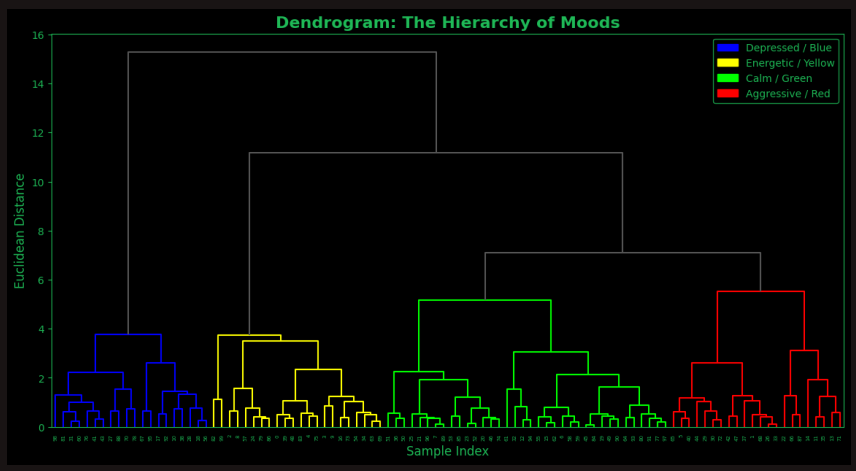

Figure 1: Hierarchical clustering dendrogram based on valence, energy, and danceability

Figure 1 illustrates the hierarchical structure of emotional similarity between tracks, constructed using **Ward's linkage method**. Ward's method merges clusters by minimizing the increase in within-cluster variance, which promotes compact and well-separated groups.

At low linkage distances, tracks with very similar emotional characteristics form small clusters. As the linkage distance increases, these clusters gradually merge into broader emotional groupings. Based on the dendrogram structure and consistency with later analyses, the tree was cut to obtain **four emotional clusters**.

To interpret these four clusters, the resulting labels were visualized in the valence–energy feature space, allowing direct observation of how emotional profiles occupy different regions of the affective plane.

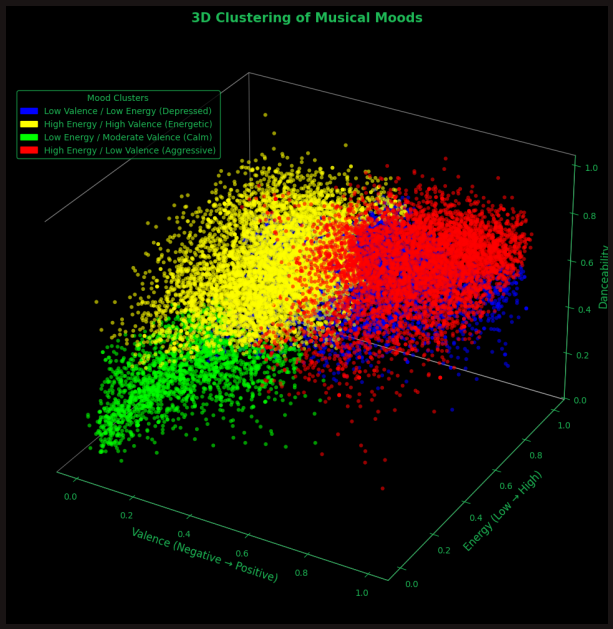

Figure 2: Four mood clusters visualized in the valence–energy feature space

As shown in Figure 2, the four clusters occupy distinct regions of the emotional feature space. Tracks with low valence and low energy form a group associated with sad or melancholic moods. High-energy and high-danceability tracks form a group corresponding to energetic or upbeat states. Tracks with lower energy but moderate to high valence tend to appear in a calm or relaxed region, while the remaining group captures emotionally mixed or transitional profiles.

Together, these visualizations show that continuous emotional variation can be organized into four interpretable emotional regions using hierarchical clustering.

3.1.4 Summary

The hierarchical clustering results indicate that four meaningful emotional profiles emerge naturally from Spotify audio features. Although musical emotion varies along a continuum, hierarchical clustering combined with visualization reveals clear and interpretable emotional regions, supporting the feasibility of mood-based music categorization.

3.2 Mood Clustering Using Gaussian Mixture Models

Gaussian Mixture Models (GMM), a Prototype method, were applied to the mood-related audio features (*valence*, *energy*, and *danceability*) to identify emotional structure while allowing soft, probabilistic membership. Unlike hierarchical clustering, GMM does not assign each track to a single rigid group, but instead estimates the probability that a track belongs to each of the four Gaussian components.

Mathematical Formulation GMM assumes that data are generated from a mixture of Gaussian distributions:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

where π_k are mixture weights, and μ_k, Σ_k are the mean and covariance of each component. Parameters are estimated using the Expectation–Maximization (EM) algorithm by maximizing

the log-likelihood:

$$\log p(X) = \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k) \right)$$

Table 1: Centroids of GMM Emotional Clusters (Standardized Features)

Cluster	Valence	Energy	Danceability
Cluster 0	-0.203	-0.367	0.208
Cluster 1	-0.117	1.065	-0.390
Cluster 2	-1.083	-1.186	-0.990
Cluster 3	1.148	0.339	0.681

Table 1 reports the centroid values of the four Gaussian components, expressed in standardized feature space. Each centroid represents the average emotional profile of the tracks most strongly associated with that component.

Cluster 2 exhibits very low valence, energy, and danceability, corresponding to melancholic or depressed moods. Cluster 3 shows high valence and high danceability with moderate energy, representing positive and upbeat emotional states. Cluster 1 is characterized by high energy but lower danceability and near-neutral valence, reflecting intense or aggressive moods. Cluster 0 displays slightly below-average valence and energy with moderate danceability, corresponding to more subdued or introspective emotional profiles.

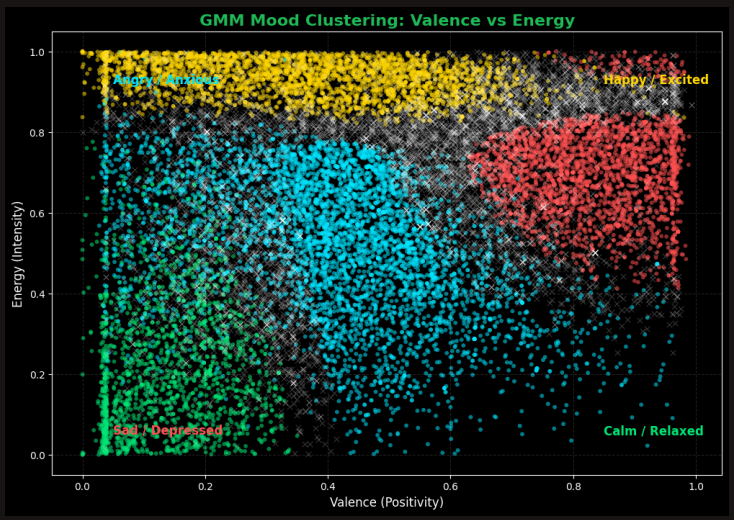

Figure 3: Gaussian Mixture Model mood clusters in the emotional feature space. Overlapping regions indicate tracks with ambiguous or hybrid emotional characteristics.

Figure 3 shows how the four Gaussian components overlap within the emotional feature space. Unlike hierarchical clustering, which produces hard assignments, GMM reveals Fuzzy transitions between moods. Tracks located near the boundaries of components exhibit lower assignment certainty, highlighting emotionally hybrid songs that share characteristics of multiple emotional profiles.

3.3 Rhythm Clustering: Identifying Groove Pockets and Outliers

Rhythm is a core element of musical structure and strongly influences how songs are perceived, danced to, and grouped into genres. While many tracks follow common rhythmic patterns,

others intentionally deviate through unusual tempos, low danceability, or irregular energy levels. This analysis investigates whether dense rhythmic patterns and rhythmically unconventional tracks can be identified using audio features alone.

Can we identify common rhythmic "groove pockets" that define standard musical patterns, while detecting rhythmic outliers that deviate from conventional structures?

3.3.1 Feature Selection

Rhythmic structure was represented using three audio features:

- **Tempo**: captures the speed of the track in beats per minute (BPM)
- **Danceability**: reflects rhythmic stability and suitability for movement
- **Energy**: represents the intensity and drive of the rhythm

Together, these features describe how fast, stable, and forceful a track's rhythm feels, making them well-suited for rhythm-focused clustering.

3.3.2 Clustering Method: DBSCAN

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) was selected for rhythm analysis because it identifies dense regions in the feature space without requiring a predefined number of clusters. Importantly, DBSCAN explicitly labels isolated points as noise, making it particularly effective for detecting rhythmically unusual or experimental tracks.

Mathematical Formulation DBSCAN defines a point as a core point if at least $MinPts$ points lie within distance ε:

$$|N_\varepsilon(x)| \geq MinPts$$

Clusters are formed by connecting density-reachable points. Points that are not reachable from any core point are labeled as noise.

3.3.3 Parameter Selection Using k-Distance Plot

To determine an appropriate value for the neighborhood radius (ε), a k-distance plot was constructed using the distance to each point's 15th nearest neighbor.

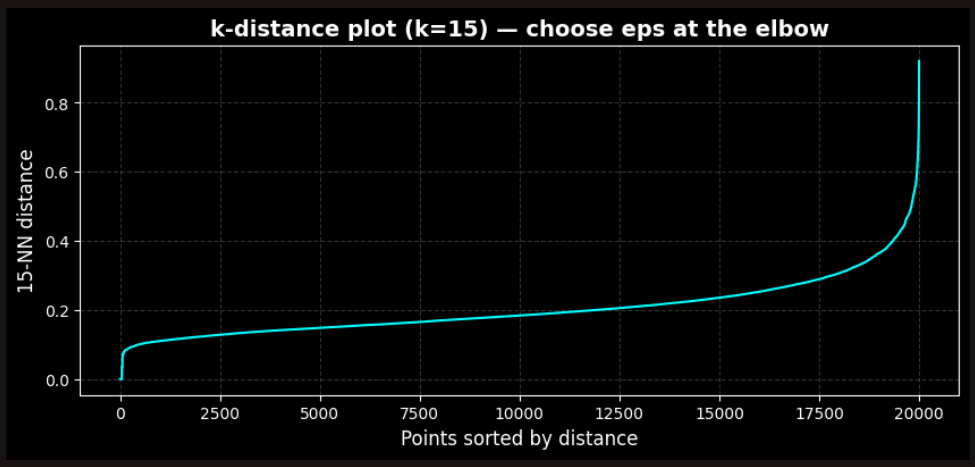

Figure 4: k-distance plot (k = 15) used to select the ε parameter for DBSCAN. The elbow point indicates a suitable threshold separating dense regions from noise.

As shown in Figure 4, the curve remains relatively flat for most points and then rises sharply. This sharp change indicates the transition from dense regions to sparse ones. The elbow point was therefore selected as the ε value, allowing DBSCAN to capture dense rhythmic patterns while excluding rhythmically atypical tracks as outliers.

3.3.4 Visualization of Rhythm Clusters

The resulting DBSCAN clusters were visualized to examine rhythmic structure and outlier behavior.

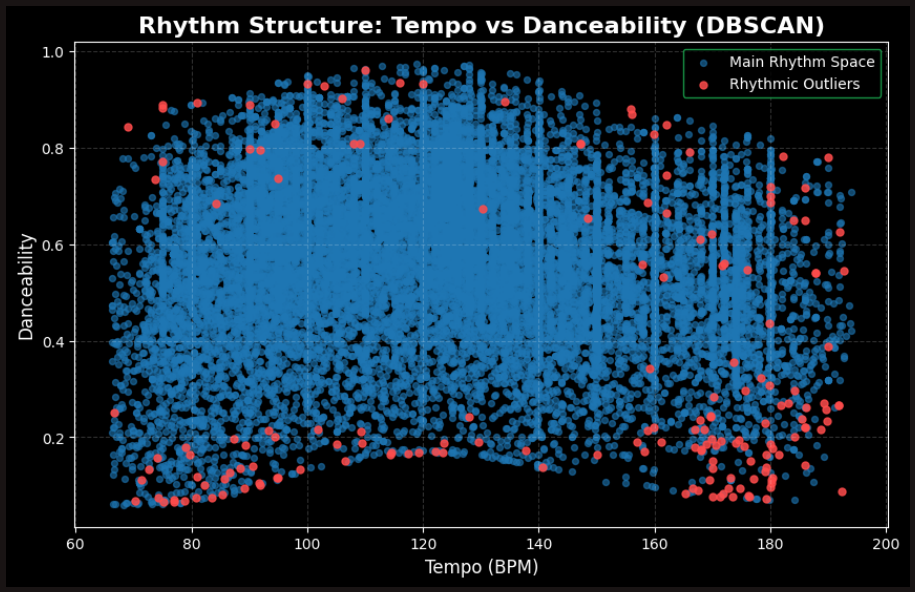

Figure 5: DBSCAN clustering visualized in the tempo–danceability space. Dense regions represent common rhythmic grooves, while isolated points are labeled as rhythmic outliers.

Figure 5 shows that most tracks form dense regions within moderate tempo and danceability ranges, corresponding to standard rhythmic grooves commonly found across genres. Outliers tend to appear at extreme tempo values or very low danceability levels, indicating unconventional rhythmic structures.

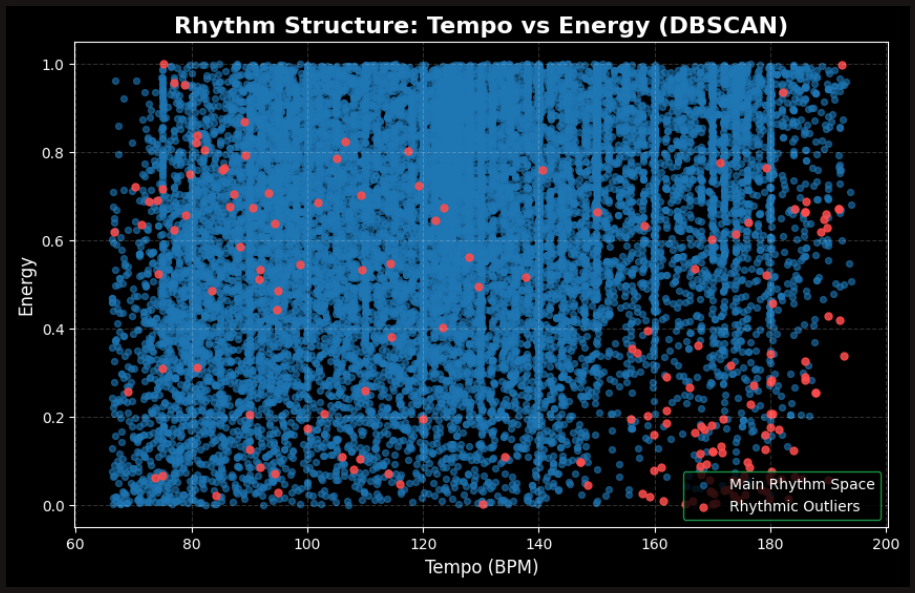

Figure 6: DBSCAN clustering visualized in the tempo–energy space. Outliers exhibit unusual combinations of tempo and energy compared to dense rhythmic regions.

Figure 6 further highlights rhythmically atypical tracks. While most songs occupy dense regions with balanced tempo and energy, outliers appear at extreme tempos or exhibit unusually low or high energy, suggesting experimental or non-standard rhythmic designs.

3.3.5 Summary

The DBSCAN-based rhythm analysis successfully identified dense rhythmic regions that define common groove patterns, while simultaneously isolating rhythmically unconventional tracks as noise. These outliers often represent experimental, genre-blending, or non-dance-oriented music that does not conform to dominant rhythmic norms.

Overall, this analysis demonstrates that rhythm-related audio features contain sufficient structure to distinguish standard groove pockets from rhythmic outliers, highlighting the effectiveness of density-based clustering for rhythm-focused music analysis.

3.4 Texture Clustering: Identifying Sonic Environments

Beyond mood and rhythm, music can also be described by its **sonic texture**, which reflects how tracks are recorded, produced, and processed. While some songs emphasize organic, acoustic instrumentation and natural dynamics, others rely heavily on electronic synthesis, studio effects, and dynamic compression.

This analysis addresses the following research question:

> *Can audio texture features be used to identify distinct sonic environments, separating organic, acoustically-driven recordings from electronically produced and highly processed tracks?*

3.4.1 Feature Selection

Three Spotify audio features were selected to represent musical texture:

- **Acousticness**: likelihood that a track is primarily acoustic
- **Instrumentalness**: degree to which a track lacks vocals

- **Loudness**: overall production intensity and compression

Together, these features capture both the sound sources and the production style of a track.

3.4.2 Clustering Method

K-Means clustering was applied to standardized texture features to ensure that all variables contributed equally to the distance calculations. The optimal number of clusters was determined using both the **elbow method** and the **silhouette score**.

Mathematical Formulation K-Means minimizes within-cluster variance:

$$\min \sum_{i=1}^{K} \sum_{x \in C_i} \|x - \mu_i\|^2$$

where C_i is cluster i and μ_i is its centroid. The algorithm alternates between assigning points to the nearest centroid and updating centroids as cluster means.

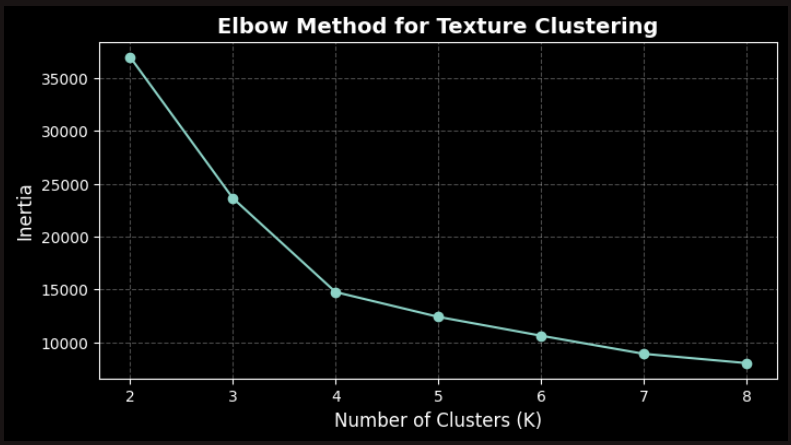

Figure 7: Elbow method for texture clustering. The curve flattens after $K = 4$, indicating diminishing returns for larger values of K.

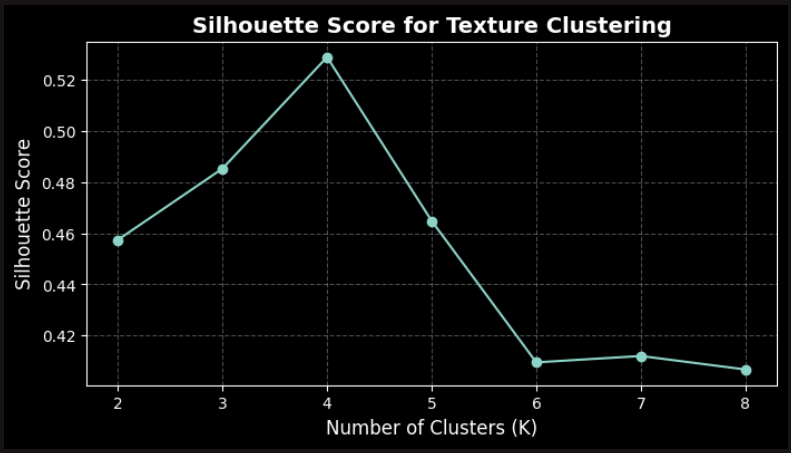

Figure 8: Silhouette score for texture clustering. The highest score occurs at $K = 4$, indicating the best balance between cluster cohesion and separation.

Both diagnostics consistently indicate $K = 4$ as the most appropriate choice, balancing compactness, separation quality, and interpretability.

3.4.3 Cluster Interpretation

Inspection of cluster centroids and feature distributions revealed four distinct sonic environments:

- **Electronic / Processed**: very high loudness and low acousticness, characteristic of heavily produced electronic music
- **Acoustic / Organic**: high acousticness and lower loudness, corresponding to unplugged or naturally recorded tracks
- **Hybrid (Vocal-Oriented)**: moderate loudness and acousticness, reflecting produced tracks with prominent vocals
- **Hybrid (Instrumental / Ambient)**: intermediate production levels with higher instrumentalness, capturing atmospheric or instrumental music

Rather than forming a strict acoustic–electronic binary, the data reveals a continuum of production styles with meaningful intermediate regions.

3.4.4 Visualization of Texture Profiles

To examine how the identified clusters differ in terms of sonic texture, the distribution of texture features across sonic environments was visualized using boxplots.

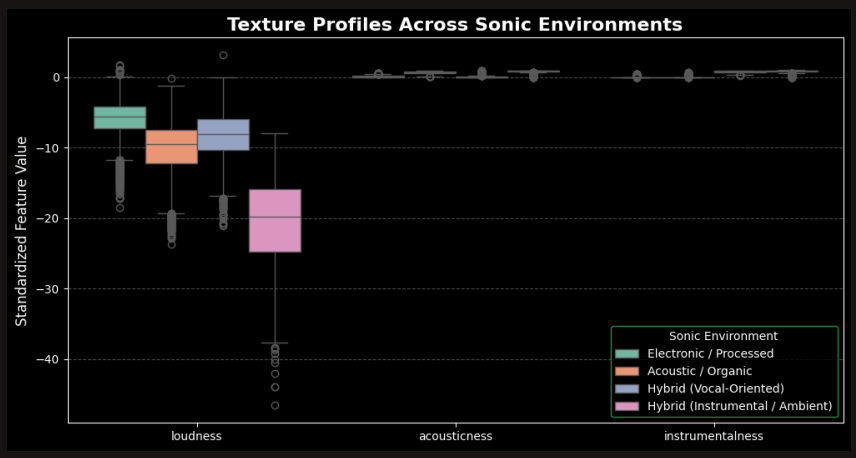

Figure 9: Distribution of standardized texture features across sonic environments. Acousticness and loudness provide the strongest separation, while instrumentalness refines hybrid regions.

Figure 9 shows a clear separation between sonic environments along the acousticness and loudness dimensions. Tracks in the *Electronic / Processed* cluster exhibit the highest loudness and lowest acousticness, reflecting heavily produced music. In contrast, the *Acoustic / Organic* cluster displays high acousticness and lower loudness, corresponding to naturally recorded tracks.

The two hybrid clusters occupy intermediate regions. The *Hybrid (Vocal-Oriented)* cluster shows moderate loudness and acousticness, while the *Hybrid (Instrumental / Ambient)* cluster is characterized by substantially lower loudness and higher instrumentalness. Instrumentalness therefore plays a key role in distinguishing between the two hybrid sonic environments.

3.4.5 Non-linear Validation via Spectral Clustering

To test whether the separation between sonic textures extends beyond linear decision boundaries, **Spectral Clustering** was applied as a complementary non-linear method. Due to the computational cost of spectral methods, this analysis was conducted on a representative subsample of the dataset.

Mathematical Formulation Spectral clustering constructs a similarity graph with weight matrix W, and degree matrix D, then forms the graph Laplacian:

$$L = D - W$$

Clustering is performed on the first K eigenvectors of L, which embed data into a space where non-linear structure becomes linearly separable.

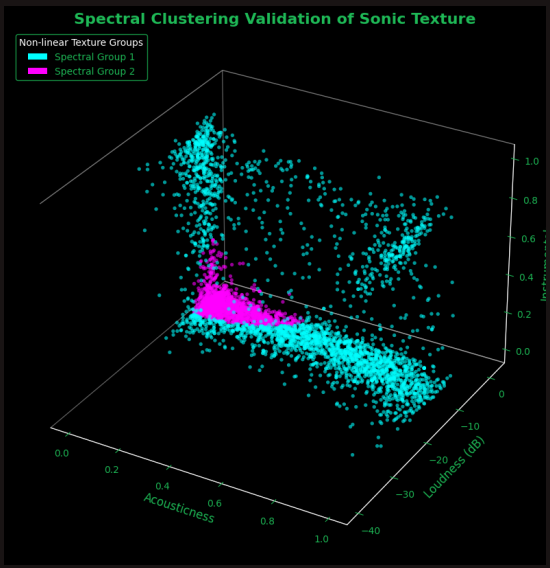

Figure 10: Spectral clustering validation of sonic texture separation in acousticness–loudness–instrumentalness space. The curved boundary highlights a non-linear split between organic and processed texture regions.

Figure 10 reveals that tracks separate into two broad texture regions under a non-linear clustering model. One region is concentrated at low acousticness and high loudness, corresponding to heavily processed music, while the other occupies higher acousticness and lower loudness values, reflecting more organic recordings. The boundary between these regions is curved rather than linear, indicating that texture separation follows a non-linear structure in the feature space.

This result supports the robustness of the texture-based structure identified by K-Means, while also showing that at a higher level, sonic texture can be viewed as a continuum between organic and processed production styles.

3.4.6 Evaluation Metrics

To provide a quantitative reference for clustering quality, the **silhouette score** was computed for each main clustering task. These values are reported for diagnostic purposes and should not be interpreted as absolute performance measures, especially when comparing different clustering paradigms.

Table 2: Silhouette Scores Across Clustering Tasks

Clustering Task	Silhouette Score
Mood Clustering (Hierarchical)	0.249
Rhythm Clustering (DBSCAN)	Not defined (single cluster)
Texture Clustering (K-Means)	0.529

The relatively higher silhouette score for texture clustering indicates clearer separation in production-related features compared to mood or rhythm, which are inherently more continuous and overlapping musical dimensions.

3.4.7 Summary

The texture clustering analysis demonstrates that Spotify audio features contain sufficient information to distinguish multiple sonic environments related to production style. The results reveal both a strong separation between organic and processed music and meaningful hybrid regions that lie between these extremes. The consistency between linear (K-Means) and non-linear (Spectral Clustering) methods further supports the robustness of the identified texture structure and reflects the blended and evolving nature of modern music production.

3.5 Subspace Clustering: Do Clusters Exist Only in Specific Dimensions?

In high-dimensional data, meaningful structure may exist only within specific subsets of features and can weaken or disappear when additional dimensions are introduced. This phenomenon is especially relevant in music analysis, where rhythmic characteristics may form clear patterns, while emotional dimensions such as valence introduce more continuous and diffuse variation.

This analysis addresses the following research question:

Do meaningful clusters emerge only in specific feature subspaces, such as tempo and energy, and become less distinct when additional dimensions like valence are considered?

To investigate this question, the grid-based subspace clustering algorithm **CLIQUE** was applied to the three-dimensional feature space defined by *tempo*, *energy*, and *valence*. CLIQUE partitions each dimension into a fixed grid and identifies dense regions independently across subspaces, making it well suited for detecting localized structure that may not persist in higher-dimensional representations.

3.5.1 Methodology

The analysis was performed on standardized feature values to ensure comparable scales across dimensions. The feature space was divided into a fixed number of intervals per dimension, and a relatively high density threshold was selected in order to capture only robust, high-density regions. Points that did not belong to any dense grid cell were labeled as noise.

Mathematical Formulation Each dimension is divided into m intervals, forming grid cells. A cell is dense if:

$$|\text{points in cell}| \geq \tau$$

where τ is the density threshold. Dense cells are combined across subspaces to form clusters, while points outside dense regions are labeled as noise.

3.5.2 CLIQUE Results

CLIQUE identified multiple dense regions in the Tempo–Energy–Valence space, along with a large proportion of points classified as noise. A summary of the clustering outcome is reported in Table 3.

Table 3: Summary of CLIQUE Subspace Clustering Results

Metric	Value
Number of dense subspace clusters	4
Total number of noise points	8104
Cluster sizes	150, 11679, 36, 31

The presence of one large dense region and several much smaller clusters indicates that most tracks share broadly similar tempo–energy characteristics, while only a small number form highly concentrated, specialized rhythmic patterns.

3.5.3 Tempo–Energy Projection

To interpret the subspace structure identified by CLIQUE, the dense regions were visualized in the Tempo–Energy projection, as shown in Figure 11. Points classified as noise appear in the background, while dense regions are highlighted.

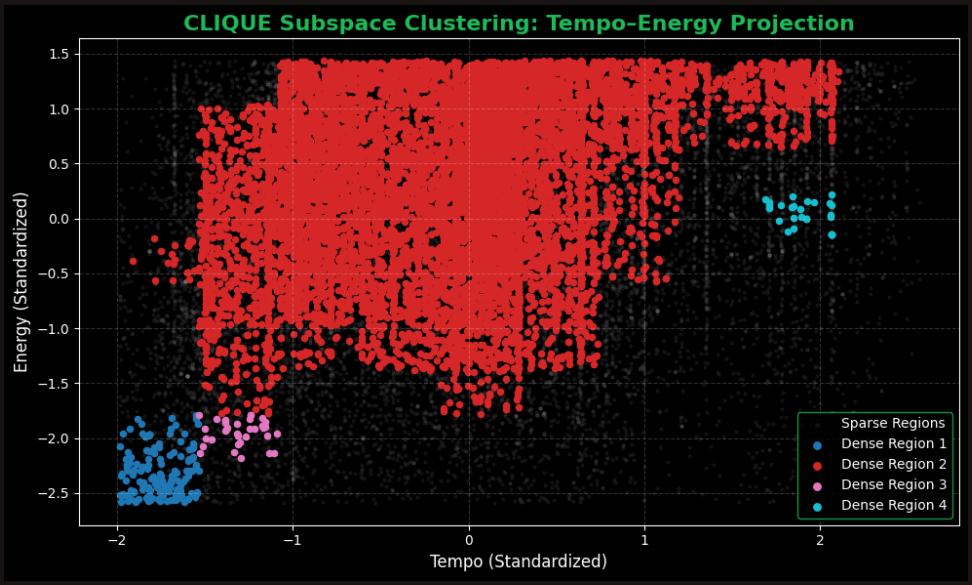

Figure 11: CLIQUE subspace clustering results projected onto the Tempo–Energy plane. Dense regions reveal clear rhythmic structure, while many points are classified as sparse when all three dimensions are considered.

Figure 11 shows that dense structure is clearly visible in the Tempo–Energy subspace. Several compact regions emerge at extreme combinations of tempo and energy, while a large central region captures the dominant rhythmic patterns present across much of the dataset. In contrast, the large number of noise points indicates that these structures become less coherent when the valence dimension is added.

3.5.4 Interpretation

These results support the hypothesis that meaningful clustering structure is strongest in rhythm-related dimensions. Tempo and energy form well-defined dense regions, while the inclusion of valence introduces additional variability that weakens density-based coherence in the three-dimensional space.

This does not imply that meaningful structure is entirely absent in higher dimensions, but rather that certain types of structure—especially dense, compact regions—become harder to detect as dimensionality increases. This highlights the importance of subspace-aware methods when analyzing complex musical data, since relevant patterns may exist only in specific feature combinations and can be obscured when too many dimensions are considered simultaneously.

3.5.5 Summary

The CLIQUE analysis demonstrates that musical clusters can be strong in specific feature subspaces while becoming less distinct when additional dimensions are introduced. In this case, rhythmic features exhibit clear density-based structure, whereas mood-related variation diffuses cluster boundaries. These findings emphasize the value of subspace clustering for uncovering localized patterns that global clustering methods may fail to capture.

3.6 Non-Linear Mood Geometry: Beyond Spherical Clusters

Emotional structure in music is often conceptualized as continuous rather than discrete. In psychology, the *Circumplex Model of Affect* represents emotions in a two-dimensional space defined by valence (positivity) and arousal (energy), where emotional states are arranged along a circular continuum. This model suggests that emotional similarity may follow curved or cyclic patterns rather than forming compact, spherical groups.

This analysis addresses the following research question:

> *Can complex, non-spherical emotional structures—such as curved or ring-like patterns—be identified in the valence–energy space using non-linear clustering methods?*

3.6.1 Methodology

To investigate non-linear structure in emotional space, **Spectral Clustering** was applied to the two-dimensional feature space defined by *valence* and *energy*. Unlike centroid-based methods such as K-Means, spectral clustering uses a graph-based representation of local similarities, allowing it to capture curved and manifold-like structures.

Due to the quadratic memory complexity of spectral methods, the analysis was performed on a random subsample of 5,000 tracks. Standardized feature values were used for clustering, while original feature scales were retained for visualization and interpretability.

3.6.2 Results and Visualization

The resulting clustering was visualized in the valence–energy plane, as shown in Figure 12.

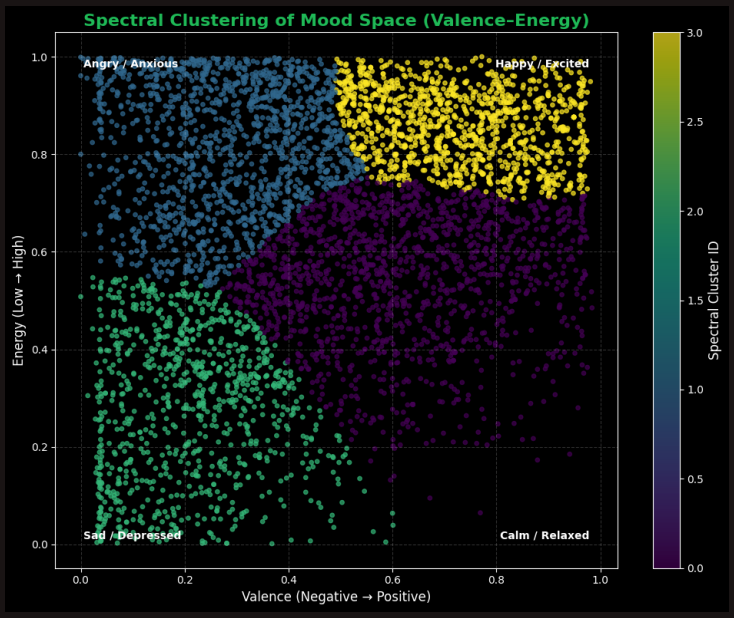

Figure 12: Spectral clustering of the valence–energy mood space. Curved, non-linear boundaries emerge between emotional regions, consistent with continuous affective transitions rather than spherical clusters.

Figure 12 shows that emotional regions are separated by curved boundaries that cannot be captured by centroid-based clustering. While no perfectly ring-shaped cluster is observed, the partitions follow smooth, curved trajectories that align with the conceptual structure of the circumplex model. Emotional regions transition gradually across valence and energy dimensions, reflecting continuous affective change rather than sharply separated categories.

3.6.3 Interpretation

The absence of a perfect circular cluster does not contradict the circumplex model. Instead, it reflects the empirical distribution of real-world music data, which is uneven and concentrated in certain emotional regions rather than uniformly filling the affective space. Spectral clustering successfully captures this non-uniform, non-linear organization, demonstrating that emotional structure exists on a continuous manifold rather than as isolated spherical clusters.

3.6.4 Summary

This analysis demonstrates that mood-related audio features exhibit complex, non-linear structure in the valence–energy space. Spectral clustering reveals curved emotional boundaries consistent with psychological theories of affect, highlighting the limitations of centroid-based clustering methods and emphasizing the importance of non-linear approaches for modeling emotional dimensions in music.

3.7 Hit Potential Across Mood Clusters

Beyond uncovering latent emotional structure in music, an important practical question is whether certain musical moods are associated with greater commercial success. Spotify provides a popularity score for each track, reflecting overall listener engagement and streaming impact.

This analysis investigates whether differences in **hit potential** emerge across mood-based clusters identified using spectral clustering. Importantly, popularity was *not* used during the clustering process, ensuring that any observed differences arise from emotional characteristics rather than information leakage.

Do certain musical mood clusters exhibit higher hit potential, as measured by Spotify popularity scores?

To address this question, Spotify popularity scores were analyzed **post hoc** across the four mood regions derived from valence and energy: *Angry / High Energy*, *Calm / Positive*, *Sad / Low Energy*, and *Happy / Energetic*. Boxplots were used to visualize the distribution of popularity within each mood region, as shown in Figure 13.

Figure 13: Popularity distribution across mood regions derived from spectral clustering. Boxes represent interquartile ranges with median popularity shown as horizontal lines.

Figure 13 shows noticeable differences in median popularity across mood regions. Tracks belonging to the *Angry / High Energy* and *Calm / Positive* clusters exhibit slightly higher median popularity scores, suggesting stronger hit potential. In contrast, the *Sad / Low Energy* and *Happy / Energetic* clusters show lower median values, although substantial overlap remains across all distributions.

These overlaps reflect the inherently noisy and long-tailed nature of Spotify popularity, which is influenced by many factors beyond audio characteristics, such as marketing, artist visibility, and listener trends. Nevertheless, the systematic shifts in median values suggest that emotional intensity and emotional stability are modestly associated with higher commercial impact.

To contextualize these results, Table 4 reports the number of tracks assigned to each mood region. The relatively balanced cluster sizes indicate that the observed popularity differences are unlikely to be driven by extreme class imbalance.

Table 4: Number of tracks per mood region in the spectral clustering subsample

Mood Cluster	**Number of Tracks**
Calm / Positive	1499
Angry / High Energy	1433
Happy / Energetic	1152
Sad / Low Energy	916

Overall, this analysis suggests that musical mood is meaningfully related to hit potential. While popularity cannot be predicted from audio features alone, certain emotional profiles—particularly emotionally intense or emotionally grounded tracks—are more frequently associated with higher popularity outcomes.

3.8 Prototype Songs: Interpreting Mood Clusters Through Representative Tracks

While clustering reveals latent structure in musical feature space, clusters remain abstract without concrete musical examples. To improve interpretability, this analysis identifies **representative prototype songs** for each mood cluster and examines their audio feature profiles.

Rather than relying on synthetic centroids, representative tracks were selected by identifying songs closest to each cluster centroid in the *valence–energy* space. These prototype songs serve as musically meaningful exemplars of each cluster, allowing qualitative interpretation of the discovered mood regions.

What audio characteristics define each mood cluster, and how can representative songs help interpret the emotional structure uncovered by clustering?

For each cluster, multiple representative tracks were selected and their audio features were analyzed using raw Spotify feature values. Figure 14 visualizes the average feature profiles of these prototype songs across five key dimensions: valence, energy, danceability, acousticness, and instrumentalness.

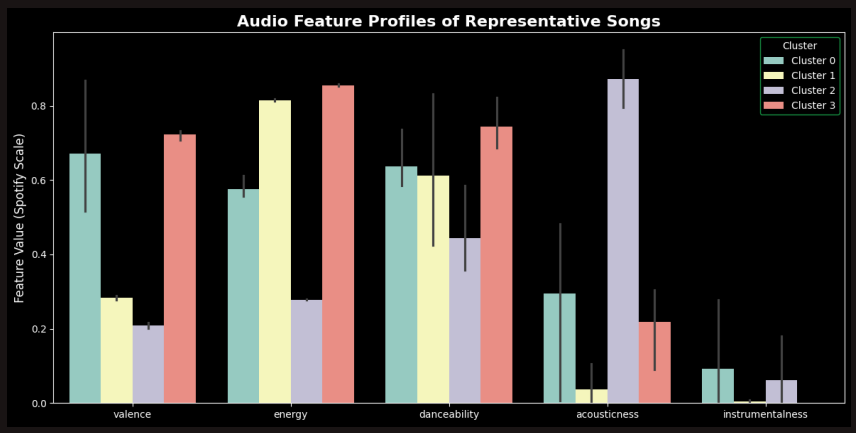

Figure 14: Audio feature profiles of representative prototype songs for each mood cluster. Feature values are shown on the original Spotify scale. Error bars indicate variability among prototype tracks within each cluster.

Figure 14 highlights clear and interpretable differences between mood clusters. One cluster is characterized by high energy, high valence, and strong danceability, corresponding to emotionally intense and rhythmically driven music. Another cluster exhibits lower energy and valence alongside elevated acousticness, reflecting more subdued and organic musical styles. Additional clusters occupy intermediate regions, blending emotional positivity with moderate rhythmic intensity or emphasizing acoustic textures over dance-oriented structure.

The acousticness and instrumentalness dimensions further refine these distinctions, separating electronically produced, vocal-driven tracks from more acoustic or instrumental compositions. Importantly, these differences emerge despite prototype selection being based solely on emotional features, demonstrating that mood-based clustering captures broader aspects of musical structure.

Overall, the use of representative prototype songs bridges the gap between abstract clustering results and intuitive musical understanding. By grounding clusters in concrete examples and interpretable feature profiles, this analysis confirms that the identified mood regions correspond to coherent and musically meaningful sound characteristics rather than arbitrary mathematical groupings. meaningful sound characteristics.

3.9 Outlier Worlds: Rare and Extreme Songs Beyond Mainstream Structure

While most clustering analyses focus on identifying dominant musical patterns, an equally important perspective is to examine tracks that do not conform to mainstream structure. Such songs may represent experimental, niche, or genre-blending music that occupies sparse regions of the feature space rather than dense musical norms.

This analysis explores rare and extreme songs using a density-based clustering approach that explicitly models noise.

What kinds of songs do not conform to dominant musical structures and emotional norms?

To address this question, the HDBSCAN algorithm was applied to a high-dimensional audio feature space capturing rhythm, emotion, and texture. Unlike centroid-based methods, HDBSCAN identifies dense clusters while labeling points in low-density regions as outliers. These outliers are not treated as erroneous data points, but as tracks exhibiting rare or extreme combinations of musical characteristics.

3.9.1 Outlier Detection and Distribution

Using conservative density parameters, a small subset of tracks was classified as outliers. These tracks lie outside the dominant musical structures learned by the clustering model.

Figure 15 visualizes the distribution of rare songs in the valence–energy space. The background density represents the overall song population, while outliers are highlighted. The visualization shows that rare tracks tend to populate the edges of the emotional space, occupying extreme low or high valence and energy regions rather than central, densely populated areas.

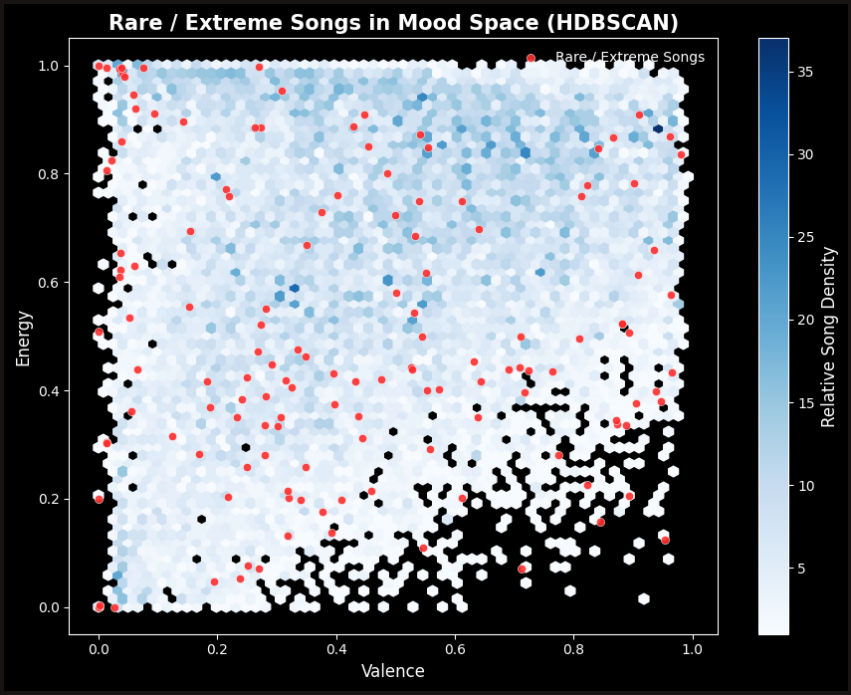

Figure 15: Rare and extreme songs identified by HDBSCAN projected onto the valence–energy mood space. Background density indicates common musical regions, while highlighted points represent low-density, non-mainstream tracks.

3.9.2 Statistical Profile of Rare Songs

To better understand the characteristics of these non-mainstream tracks, descriptive statistics were computed across key audio features. Table 5 summarizes the distribution of rhythmic, emotional, and textural features among the outlier songs.

Table 5: Descriptive statistics of rare and extreme songs identified as outliers by HDBSCAN

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Tempo	139	127.61	38.59	67.32	91.42	125.72	168.42	193.98
Energy	139	0.51	0.27	0.00	0.33	0.44	0.75	1.00
Valence	139	0.43	0.30	0.00	0.20	0.38	0.64	0.98
Danceability	139	0.56	0.22	0.06	0.45	0.61	0.71	0.97
Acousticness	139	0.52	0.35	0.00	0.15	0.54	0.85	1.00
Instrumentalness	139	0.38	0.39	0.00	0.00	0.30	0.81	1.00
Speechiness	139	0.45	0.23	0.03	0.29	0.43	0.60	0.96

The statistics show that rare songs exhibit wide variability across nearly all dimensions. Tempo spans from slow to extremely fast, while energy and valence range from near-zero to maximal values, indicating emotional extremes. Elevated instrumentalness and speechiness suggest that many outliers correspond to instrumental pieces, spoken-word tracks, or unconventional vocal structures. The broad acousticness distribution further reflects both highly produced electronic tracks and minimally processed acoustic recordings.

3.9.3 Interpretation

These findings indicate that musical outliers are not random anomalies, but structurally distinct tracks that deviate from dominant musical norms along multiple dimensions. They often occupy extreme emotional states, unconventional rhythmic regimes, or atypical production styles. As such, they represent creative, experimental, or niche musical worlds that are poorly captured by global clustering methods but naturally revealed through density-based analysis.

3.9.4 Summary

This analysis demonstrates the effectiveness of HDBSCAN for identifying rare and extreme musical structures. By explicitly modeling low-density regions as meaningful deviations rather than mere noise, the method uncovers a diverse set of non-mainstream tracks that expand the musical landscape beyond dominant mood, rhythm, and texture patterns.

4 Results

Across all analyses, the clustering methods revealed clear and interpretable structure in Spotify audio features. Mood-related dimensions formed meaningful emotional regions, rhythm-related features produced dense groove patterns with identifiable outliers, and texture features separated organic and processed sonic environments. Subspace and non-linear analyses further showed that musical structure is not uniform across all dimensions, but emerges most strongly in specific feature combinations.

Together, these results demonstrate that unsupervised learning can uncover musically meaningful patterns without relying on genre labels or manual annotation.

5 Discussion

The findings of this project align well with intuitive musical concepts such as mood, groove, and production style, while also revealing more subtle structure that is not immediately visible through genre labels alone. Hierarchical clustering provided an interpretable view of how emotional similarity gradually merges across songs, while Gaussian Mixture Models captured soft transitions between emotional states. Density-based methods such as DBSCAN and HDBSCAN proved especially effective at identifying unconventional or experimental tracks that fall outside dominant musical norms.

The combination of linear, non-linear, and subspace-based methods highlighted an important insight: musical structure is not captured by a single "best" clustering algorithm. Instead, different methods reveal different aspects of musical organization. This reinforces the idea that music is multi-dimensional and continuous, and that no single model can fully describe its complexity.

6 Limitations

- The absence of ground-truth mood labels limits the ability to perform formal quantitative validation of clustering quality.
- Cluster structure depends on feature selection and preprocessing choices, which may influence the patterns that emerge.
- Musical perception is shaped by culture, language, and personal experience, none of which are captured by audio features alone.
- Popularity is influenced by many external factors such as marketing and artist visibility, which are not represented in the dataset.

7 Conclusion

This project demonstrates that Spotify audio features contain rich information about musical emotion, rhythm, and production style, and that unsupervised learning methods can uncover meaningful structure within this space. By combining multiple clustering approaches, the analysis revealed emotional regions, groove patterns, sonic environments, and rare musical outliers without relying on genre labels.

Beyond technical results, this work shows that data-driven methods can support more intuitive ways of exploring music — from mood-based playlist generation to discovery of unconventional or niche tracks. While music will always involve subjective and cultural interpretation, clustering provides a powerful tool for organizing and navigating large musical collections in a more human-centered way.

References

- Spotify Audio Features Documentation.
- Pedregosa et al., *Scikit-learn: Machine Learning in Python.*